

Causal Inference

Lecture 10

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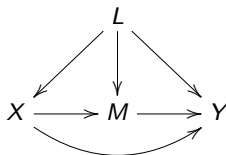
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Notations

- Assume a treatment X , an outcome Y , a mediator M , and measured covariates L .



DAG for mediation analysis

Motivating Example

Education setting

- Intervention X : a new teaching method in the classroom.
- Outcome Y : students' end-of-semester exam performance.
- Mediator M : students' engagement or interest in the subject.
- Measured covariates L : students' baseline aptitude and study habits.

Why consider mediation?

- The new teaching method might raise exam performance partly by directly improving teaching quality.
- It might also indirectly increase performance by boosting interest in the subject, which then leads to higher engagement and better scores.
- We want to distinguish the direct effect of the teaching method from the portion (the indirect effect) that is mediated by engagement and interest in the subject.

Nested Potential Outcomes

Potential outcomes

- Assume a binary outcome X . Let the intervention be such that X is set x , 0, 1, then the potential outcome and the potential mediator are

$$Y_x, M_x, \quad x = 0, 1.$$

- Assume now that we intervene both on the exposure X and the outcome M , and set them to x and m , respectively. Thus, the potential outcomes are

$$Y_{x,m}, \quad x = 0, 1; \quad m \in \mathcal{M}$$

Nested potential outcomes

- Assume that we set the exposure X is to x for the outcome Y , and X to x' for the mediator. Hence, the nested potential outcome is defined as

$$Y_{x,M_{x'}}, \quad x \text{ and } x' = 0, 1.$$

where x' can be different from x .

Composition - one world

- The notation $Y_{x,M_{x'}}$ is the hypothetical outcome if the treatment X were set at level x and the mediator were set at its potential level $M_{x'}$ under treatment $X = x'$.

$$Y_{x,M_{x'}}, \quad x \text{ and } x' = 0, 1.$$

where x' can be different from x .

- With a binary exposure X , we have four nested potential outcomes in total:

$$Y_{0,M_0}, Y_{0,M_1}, Y_{1,M_0}, Y_{1,M_1}.$$

- Assumption (composition):

$$Y_{x,M_x} = Y_x, \quad x \in \mathcal{X}.$$

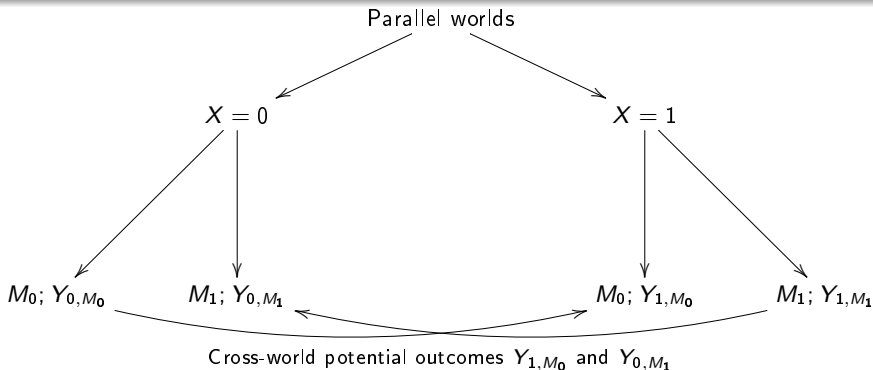
Namely, $Y_{1,M_1} = Y_1$ and $Y_{0,M_0} = Y_0$. That is, for every subject we observe wither Y_{1,M_1} or Y_{0,M_0}

Cross-world intervention

- The nested potential outcome $Y_{x,M_{x'}}$, where $x \neq x'$ is the hypothetical outcome if the subject received treatment x but his mediator were set at its natural value $M_{x'}$ under the exposure $X = x'$.
- Similarly, Y_{x',M_x} is the hypothetical outcome if the subject received control x' but its mediator were set at its natural value M_x under the exposure $X = x$. These are two *cross-world* counterfactual terms and are useful for defining the direct and indirect effects.
- For a binary exposure $X = 0, 1$, Y_{0,M_1} and Y_{1,M_0} are, by definition, not observable quantities in any experiments. Therefore, generally, we cannot falsify or check any statements involving these quantities.
- Even if we randomize exposure and the mediator, we still cannot observe a subject that has exposure x , and the mediator value under exposure x' .

Cross-world intervention

- The nested counterfactuals Y_{0,M_1} and Y_{1,M_0} are, by definition, not observable quantities in any experiments since they involve quantities from distinct and exclusive realizations (worlds).
- Therefore, generally, we cannot falsify or check any statements involving these quantities.



Fill in the missing values

ID	M_0	M_1	Y_{00}	Y_{01}	Y_{10}	Y_{11}	Y_{0M_0}	Y_{0M_1}	Y_{1M_0}	Y_{1M_1}
1	0	0	0	0	1	1				
2	0	1	0	1	1	1				
3	1	1	0	0	0	1				

ID	M_0	M_1	Y_{00}	Y_{01}	Y_{10}	Y_{11}	Y_{0M_0}	Y_{0M_1}	Y_{1M_0}	Y_{1M_1}
1	0	0	0	0	1	1	0 (= Y_{00})	0 (= Y_{00})	1 (= Y_{10})	1 (= Y_{10})
2	0	1	0	1	1	1	0 (= Y_{00})	1 (= Y_{01})	1 (= Y_{10})	1 (= Y_{11})
3	1	1	0	0	0	1	0 (= Y_{01})	0 (= Y_{01})	1 (= Y_{11})	1 (= Y_{11})

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Definition

- For a binary exposure $X = 0, 1$, the total effect (TE) of the exposure is defined as

$$TE = \mathbb{E}[Y_1 - Y_0] = \mathbb{E}[Y_{1,M_1} - Y_{0,M_0}].$$

The total effect is the average treatment effect (ATE).

Definition

- For a binary exposure $X = 0, 1$, the natural direct effect (NDE) for $X = 1$ is defined as

$$NDE(1) = \mathbb{E}[Y_{1,M_1} - Y_{0,M_1}].$$

The NDE measures the effect of the exposure on the outcome if the mediator attains the value it would have attained if the exposure is set to 1.

- Analogously, the NDE for $X = 0$ is defined as

$$NDE(0) = \mathbb{E}[Y_{1,M_0} - Y_{0,M_0}].$$

- By holding $X = x$ fixed we ensure that the mediator does not vary due to the exposure:
- It may still vary for other reasons, e.g., because L varies.
- Hence, any variation in the outcome when X is taken from 0 to 1 must be due to a direct effect of the exposure X .

Definition

- For a binary exposure $X = 0, 1$, the natural indirect effect (NIE) for $X = 1$ is defined as

$$NIE(1) = \mathbb{E}[Y_{1,M_1} - Y_{1,M_0}].$$

- Analogously, the NDE for $X = 0$ is defined as

$$NIE(0) = \mathbb{E}[Y_{0,M_1} - Y_{0,M_0}].$$

- The NIE measures the effect transmitted *through* the mediator, contrasting the mediator's natural values under $X = 1$ versus those under $X = 0$, while holding the exposure at $X = 1$ in both terms.
- By holding X fixed we ensure that the outcome does not vary due to a direct effect of the exposure
- By taking X from 0 to 1 we let the mediator vary due to the exposure, and this variation is propagated to the outcome.
- Hence, any variation in the outcome when X is taken from 0 to 1 must be due to an indirect effect of the exposure.

Total effect decomposition

- For a binary exposure $X = 0, 1$, the natural direct and indirect effects add up to the total effect

$$TE = NDE(x) + NIE(1 - x).$$

- Proof:

$$\begin{aligned} TE &= \mathbb{E}[Y_{1,M_1} - Y_{0,M_0}] \\ &= \mathbb{E}[Y_{1,M_1} - Y_{0,M_0} - Y_{1,M_0} + Y_{1,M_0}] \\ &= \mathbb{E}[Y_{1,M_1} - Y_{1,M_0} + Y_{1,M_0} - Y_{0,M_0}] \\ &= \mathbb{E}[Y_{1,M_1} - Y_{1,M_0}] + \mathbb{E}[Y_{1,M_0} - Y_{0,M_0}] \\ &= NIE(1) + NDE(0). \end{aligned}$$

- Analogically,

$$\begin{aligned} TE &= \mathbb{E}[Y_{1,M_1} - Y_{0,M_0}] \\ &= \mathbb{E}[Y_{1,M_1} - Y_{0,M_0} - Y_{0,M_1} + Y_{0,M_1}] \\ &= \mathbb{E}[Y_{0,M_1} - Y_{0,M_0}] + \mathbb{E}[Y_{1,M_1} - Y_{0,M_1}] \\ &= NIE(0) + NDE(1). \end{aligned}$$

Use the following table to compute the $NDE(0)$, $NDE(1)$, and $NIE(0)$, and $NIE(1)$, and the TE.

ID	M_0	M_1	Y_{00}	Y_{01}	Y_{10}	Y_{11}	Y_{0M_0}	Y_{0M_1}	Y_{1M_0}	Y_{1M_1}
1	0	0	0	0	1	1	0 (= Y_{00})	0 (= Y_{00})	1 (= Y_{10})	1 (= Y_{10})
2	0	1	0	1	1	1	0 (= Y_{00})	1 (= Y_{01})	1 (= Y_{10})	1 (= Y_{11})
3	1	1	0	0	0	1	0 (= Y_{01})	0 (= Y_{01})	1 (= Y_{11})	1 (= Y_{11})

ID	M_0	M_1	Y_{00}	Y_{01}	Y_{10}	Y_{11}	Y_{0M_0}	Y_{0M_1}	Y_{1M_0}	Y_{1M_1}
1	0	0	0	0	1	1	0 (= Y_{00})	0 (= Y_{00})	1 (= Y_{10})	1 (= Y_{10})
2	0	1	0	1	1	1	0 (= Y_{00})	1 (= Y_{01})	1 (= Y_{10})	1 (= Y_{11})
3	1	1	0	0	0	1	0 (= Y_{01})	0 (= Y_{01})	1 (= Y_{11})	1 (= Y_{11})

- $NDE(0) = \mathbb{E}[Y_{1,M_0} - Y_{0,M_0}] = 3/3 - 0/3 = 3/3.$
- $NDE(1) = \mathbb{E}[Y_{1,M_1} - Y_{0,M_1}] = 3/3 - 1/3 = 2/3.$
- $NIE(0) = \mathbb{E}[Y_{0,M_1} - Y_{0,M_0}] = 1/3 - 0/3 = 1/3.$
- $NIE(1) = \mathbb{E}[Y_{1,M_1} - Y_{1,M_0}] = 3/3 - 0/3 = 0.$
- $TE = NDE(0) + NIE(1) = 1 + 0 = NDE(1) + NIE(0) = 2/3 + 1/3 = 1.$

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Sequential ignorability

- For the identification of the natural direct and indirect effects we need to assume several more assumptions.
- We require that no unmeasured confounders affect the outcome Y , the mediator M , and the exposure X , or any pair of these variables, conditioned on L .
- Formally, such assumptions can be summarized by two conditional independence statements that are known as sequential ignorability:

$$\{Y_{x',m}, M_x\} \perp\!\!\!\perp X \mid L = l, \quad Y_{x',m} \perp\!\!\!\perp M_x \mid X = x, L = l,$$

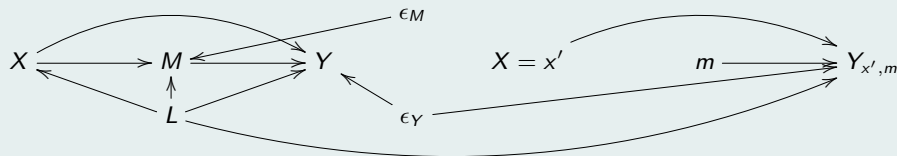
where $\mathbb{E}[X \mid L = l] > 0$ and $P(M_x = m \mid X = x, L = l) > 0$, for $x \in \{0, 1\}$.

- These assumptions are called sequential ignorability (or exchangeability) because two ignorability assumptions are made sequentially.
- The first assumption means that there are no unmeasured confounders confound the outcome or the mediator and the exposure.
- The second ignorability assumption means that there are no unmeasured confounders confound the outcome and the mediator.
- Such an assumption is not guaranteed to hold even in RCTs where the exposure (treatment) is randomized. Therefore, they depend on the subject-matter knowledge and cannot be tested using the observed data.

Decomposition of the sequential ignorability into four distinct assumptions

$$Y_{x',m} \perp\!\!\!\perp X | L$$

The exposure X and the outcome Y are unconfounded with unmeasured confounders.



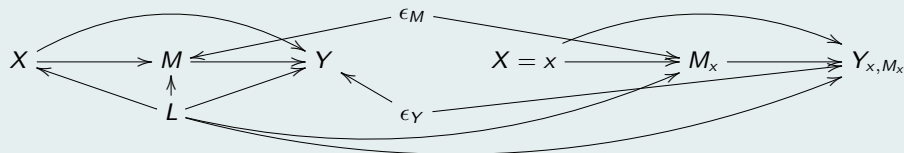
DAG of a twin network causal model with a mediator that illustrates the first part of the sequential ignorability, namely, the conditional independence of $Y_{x',m}$ and X , given L ; $Y_{x',m} \perp\!\!\!\perp X | L$.

The left-hand side of the DAG represents the observed actual world, while the right-hand side represents the hypothetical potential world.

Decomposition of the sequential ignorability into four distinct assumptions

$$M_x \perp\!\!\!\perp X | L$$

The mediator M and the exposure X are unconfounded with unmeasured confounders.

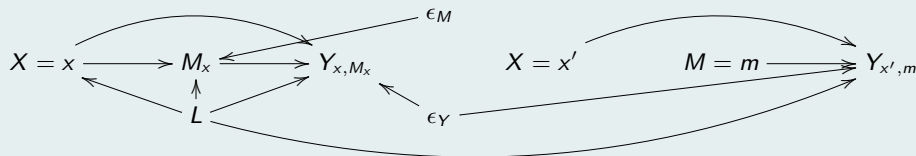


DAG of a twin network causal model with a mediator that illustrates the first part of the sequential ignorability, namely, the conditional independence of M_x and X , given L ; $M_x \perp\!\!\!\perp X | L$. The left-hand side of the DAG represents the observed actual world, while the right-hand side represents the hypothetical potential world.

Decomposition of the sequential ignorability into four distinct assumptions

$Y_{x',m} \perp\!\!\!\perp M_x | L, X = x$ - cross-world independence

There is cross-world independence between the potential outcomes $Y_{x',m}$ and potential mediators M_x for all x, x' and m .



DAG of a twin network causal model with a mediator that illustrates the second part of the sequential ignorability.

The left-hand side of the DAG represents the hypothetical world where only the exposure is set to x , while the right-hand side represents the hypothetical world where both the exposure and the mediator are set to x' and m , respectively.

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Discrete exposure, mediator, and covariate

- Pearl's mediation formula provides a non-parametric method to **identify the nested counterfactuals** using the **observed data**.
- For a discrete value exposure X , mediator M , and covariate L , then the mediation formula is given by

$$\mathbb{E}[Y_{x, M_{x'}} | L = l] = \sum_m \mathbb{E}[Y | X = x, M = m, L = l] \mathbb{P}(M = m | X = x', L = l).$$

- Therefore, the marginal expectation is given by

$$\begin{aligned} \mathbb{E}[Y_{x, M_{x'}}] &= \sum_l \mathbb{E}[Y_{x, M_{x'}} | L = l] \mathbb{P}(L = l) \\ &= \sum_l \sum_m \mathbb{E}[Y | X = x, M = m, L = l] \mathbb{P}(M = m | X = x', L = l) \mathbb{P}(L = l) \end{aligned}$$

Continuous exposure, mediator, and covariate

For a continuous value exposure X , mediator M , and covariate L , then the mediation formula is given by

$$\mathbb{E}[Y_{x,M_{x'}} | L = l] = \int_m \mathbb{E}[Y | X = x, M = m, L = l] f(M = m | X = x', L = l) dm.$$

Therefore, the marginal expectation is given by

$$\begin{aligned} \mathbb{E}[Y_{x,M_{x'}}] &= \int_l \mathbb{E}[Y_{x,M_{x'}} | L = l] f(L = l) dl \\ &= \int_l \int_m \mathbb{E}[Y | X = x, M = m, L = l] f(M = m | X = x', L = l) f(L = l) dmdl \end{aligned}$$

General form of the mediation formula

Non-parametric identification of the nested-counterfactuals using observed data

$$\mathbb{E}[Y_{x,M_{x'}} | L] = \mathbb{E}[\mathbb{E}[Y | X = x, M, L] | X = x', L].$$

Estimation

$$\mathbb{E}[Y_{x,M_{x'}}|L] = \sum_m \mathbb{E}[Y|X = x, M = m, L] \mathbb{P}(M = m|X = x', L).$$

Direct application of the mediation formula

- Fit regression models to the outcome and the mediator. Namely, for the outcome we have

$$\hat{\mathbb{E}}[Y|X = x, M = m, L] = \mathbb{E}[Y|X = x, M = m, L; \hat{\beta}],$$

and for the mediator we have

$$\hat{\mathbb{P}}(M = m|X = x', L) = \mathbb{P}(M = m|X = x', L; \hat{\gamma}).$$

- Estimate the nested counterfactual $\hat{\mathbb{E}}[Y_{x,M_{x'}}]$ using the observed data via the mediation formula

$$\hat{\mathbb{E}}[Y_{x,M_{x'}}] = \sum_m \mathbb{E}[Y|X = x, M = m, L; \hat{\beta}] \mathbb{P}(M = m|X = x', L; \hat{\gamma}).$$

From nested counterfactuals to the NDE and NIE

The identification formulas of the nested counterfactuals leads to the identification formulas (using observed data) of the natural direct and indirect effects.

Natural direct effect

Assuming sequential ignorability we have

$$\begin{aligned} NDE(0; l) &= \mathbb{E}[Y_{1,M_0} - Y_{0,M_0} | L = l] \\ &= \sum_m (\mathbb{E}[Y | X = 1, M = m, L = l] - \mathbb{E}[Y | X = 0, M = m, L = l]) \\ &\quad \times \mathbb{P}(M = m | X = 0, L = l). \end{aligned}$$

The marginal NDE is identified by averaging over the covariate L

$$NDE = \mathbb{E}[NDE(0; L)].$$

Natural indirect effect

Assuming sequential ignorability we have

$$\begin{aligned} NIE(1; l) &= \mathbb{E}[Y_{1,M_1} - Y_{1,M_0} | L = l] \\ &= \sum_m \mathbb{E}[Y | X = 1, M = m, L = l] \\ &\quad \times (\mathbb{P}(M = m | X = 1, L = l) - \mathbb{P}(M = m | X = 0, L = l)) \end{aligned}$$

The marginal NIE is identified by averaging over the covariate L

$$NIE = \mathbb{E}[NIE(1; L)].$$

Natural indirect effect with a binary mediator

- Assuming sequential ignorability and binary mediator M , the NIE is the product of the pathwise average causal effects

$$NIE(x; l) = \tau_{X \rightarrow M}(l) \tau_{M \rightarrow Y}(x, l).$$

where

$$\tau_{X \rightarrow M}(l) = \mathbb{P}(M = 1 | X = 1, L = l) - \mathbb{P}(M = 1 | X = 0, L = l)$$

and

$$\tau_{M \rightarrow Y}(x, l) = \mathbb{E}[Y = 1 | X = x, M = 1, L = l] - \mathbb{E}[Y = 1 | X = x, M = 0, L = l].$$

- Notably, $\tau_{X \rightarrow M}(l)$ can be viewed as the conditional (on L) average causal effect of the exposure X on the mediator M , and $\tau_{M \rightarrow Y}(x, l)$ as the conditional (on L and X) average causal effect of the mediator M on the outcome Y .
- The conditional (on L) NIE is the product of the aforementioned causal effect.
- This aligns with the intuition the the indirect effect acts from the exposure X to the mediator M and then from the mediator M to the outcome Y .

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Direct and Indirect effects identification and estimation in linear models

- The aforementioned identification formulas of the NDE and the NIE are non-parametric. It allows us to derive various formulas for mediation analysis under different models.
- We will start with the introduction of the linear models case, which is the most used in practice.

The moderator–mediator variable distinction in social psychological research: Conceptual, strategic, and statistical considerations.

RM Baron, [DA Kenny](#) - ... of personality and **social psychology**, 1986 - [psycnet.apa.org](#)

... the properties of **moderator** and **mediator variables** in such a ... in which **conceptual variables** may account for **differences** in ... third **variables**: (a) the **moderator** function of third **variables**, ...

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The 1986 article by Baron and Kenny on mediation analysis has accumulated 134,203 citations (January 21, 2025), making it one of the most—if not the most—cited methodological articles in history and placing it among the most highly cited academic papers overall.

The models

- Let Y be the outcome, M the discrete valued mediator, a binary exposure X , and a covariate L .
- The mediator model is:

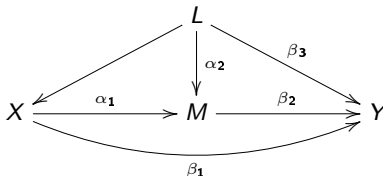
$$\mathbb{E}[M|X, L] = \alpha_0 + \alpha_1 X + \alpha_2 L$$

- The outcome model is:

$$\mathbb{E}[Y|X, M, L] = \beta_0 + \beta_1 X + \beta_2 M + \beta_3 L.$$

- The only requirement for the forthcoming results is linearity of the mediator and the outcome models.
- The specific structures of M (discrete values) and X (binary) are assumed for clarity exposition.
- Note: although both mean models are observed data models, under sequential ignorability, these models are also the structural mean model as $\mathbb{E}[M_x|L] = \mathbb{E}[M_x|X = x, L] = \mathbb{E}[M|X = x, L]$ and $\mathbb{E}[Y_{x,m}|L] = \mathbb{E}[Y_{x,m}|X = x, M = m, L] = \mathbb{E}[Y|X = x, M = m, L]$.

The Baron-Kenny formulas and DAG



DAG with coefficients for mediation analysis with linear models.

$$\mathbb{E}[M|X, L] = \alpha_0 + \alpha_1 X + \alpha_2 L$$

$$\mathbb{E}[Y|X, M, L] = \beta_0 + \beta_1 X + \beta_2 M + \beta_3 L.$$

Baron-Kenny formulas for mediation analysis

Assuming sequential ignorability and linear models as described above, we have

$$NDE = \beta_1, \quad NIE = \alpha_1 \beta_2.$$

The direct effects equals the coefficient on the path $X \rightarrow Y$, and the indirect effect equals the product of the coefficients on the path $X \rightarrow M \rightarrow Y$.

The Baron-Kenny formulas - proof for the NDE

The models are

$$\mathbb{E}[M|X, L] = \alpha_0 + \alpha_1 X + \alpha_2 L, \quad \mathbb{E}[Y|X, M, L] = \beta_0 + \beta_1 X + \beta_2 M + \beta_3 L.$$

$$\begin{aligned} NDE(0; l) &= \sum_m (\mathbb{E}[Y|X = 1, M = m, L = l] - \mathbb{E}[Y|X = 0, M = m, L = l]) \\ &\quad \times \mathbb{P}(M = m|X = 0, L = l) \\ &= \sum_m (\beta_0 + \beta_1 \times 1 + \beta_2 m + \beta_3 l - (\beta_0 + \beta_1 \times 0 + \beta_2 m + \beta_3 l)) \\ &\quad \times \mathbb{P}(M = m|X = 0, L = l) \\ &= \beta_1 \sum_m \mathbb{P}(M = m|X = 0, L = l) \\ &= \beta_1 \times 1 \\ &= \beta_1 \quad \square \end{aligned}$$

where $\sum_m \mathbb{P}(M = m|X = 0, L = l) = 1$ since it is a sum of a mass probability function of M given X and L over all its values.

Since the result is independent of $L = l$ and $X = x$, the conditional NDE equals the marginal, i.e., $NDE(x; l) = NDE$.

The Baron-Kenny formulas - proof for the NIE

The models are

$$\mathbb{E}[M|X, L] = \alpha_0 + \alpha_1 X + \alpha_2 L, \quad \mathbb{E}[Y|X, M, L] = \beta_0 + \beta_1 X + \beta_2 M + \beta_3 L.$$

$$\begin{aligned} NIE(1; l) &= \sum_m \mathbb{E}[Y|X = 1, M = m, L = l] \\ &\quad \times (\mathbb{P}(M = m|X = 1, L = l) - \mathbb{P}(M = m|X = 0, L = l)) \\ &= \sum_m (\beta_0 + \beta_1 \times 1 + \beta_2 m + \beta_3 l) \\ &\quad \times (\mathbb{P}(M = m|X = 1, L = l) - \mathbb{P}(M = m|X = 0, L = l)) \\ &= (\beta_0 + \beta_1 + \beta_3 l) \left(\sum_m \mathbb{P}(M = m|X = 1, L = l) - \sum_m \mathbb{P}(M = m|X = 0, L = l) \right) \\ &\quad + \beta_2 \left(\sum_m m \mathbb{P}(M = m|X = 1, L = l) - m \sum_m \mathbb{P}(M = m|X = 0, L = l) \right) \end{aligned}$$

The Baron-Kenny formulas - proof for the NIE: cont'd

The models are

$$\mathbb{E}[M|X, L] = \alpha_0 + \alpha_1 X + \alpha_2 L, \quad \mathbb{E}[Y|X, M, L] = \beta_0 + \beta_1 X + \beta_2 M + \beta_3 L.$$

$$\begin{aligned} NIE(1; l) &= (\beta_0 + \beta_1 + \beta_3 l) \left(\sum_m \mathbb{P}(M = m|X = 1, L = l) - \sum_m \mathbb{P}(M = m|X = 0, L = l) \right) \\ &\quad + \beta_2 \left(\sum_m m \mathbb{P}(M = m|X = 1, L = l) - \sum_m m \mathbb{P}(M = m|X = 0, L = l) \right) \\ &= (\beta_0 + \beta_1 + \beta_3 l)(1 - 1) + \beta_2(\mathbb{E}[M|X = 1, L = l] - \mathbb{E}[M|X = 0, L = l]) \\ &= \beta_2(\alpha_0 + \alpha_1 \times 1 + \alpha_2 l - \alpha_0 - \alpha_1 \times 0 - \alpha_2 l) \\ &= \beta_2 \alpha_1 \quad \square \end{aligned}$$

where $\sum_m m \mathbb{P}(M = m|X = x, L = l) = \mathbb{E}[M|X = x, L = l]$ since it is, by definition, the expected value of the mediator M given X and L .

The Baron-Kenny method - estimation and inference

The models are

$$\mathbb{E}[M|X, L] = \alpha_0 + \alpha_1 X + \alpha_2 L, \quad \mathbb{E}[Y|X, M, L] = \beta_0 + \beta_1 X + \beta_2 M + \beta_3 L.$$

OLS and Sobel test

- Under sequential ignorability, one can use the OLS method to estimate the models' coefficients $\hat{\alpha}$, $\hat{\beta}$, and then the NDE and the NIE can be estimated accordingly, i.e.,

$$\widehat{NDE} = \hat{\beta}_1, \quad \widehat{NIE} = \hat{\alpha}_1 \hat{\beta}_2.$$

- For the NDE estimator the inference is equivalent to the inference on the OLS estimator of β_1 .
- Sobel (1982, 1986) used the delta method to obtain the standard error of the NIE OLS-based estimator.
- The asymptotic variance of the NIE OLS-based estimator is

$$\text{var}(\hat{\beta}_2) \hat{\alpha}_1^2 + \hat{\beta}_2^2 \text{var}(\hat{\alpha}_1).$$

- This result allows to test the null hypothesis of no indirect effect, i.e., $H_0 : NIE = 0$.
- Most statistical software has standard procedures to perform the Sobel test.

The models are

$$\mathbb{E}[M|X, L] = \alpha_0 + \alpha_1 X + \alpha_2 L, \quad \mathbb{E}[Y|X, M, L] = \beta_0 + \beta_1 X + \beta_2 M + \beta_3 L.$$

Sobel test computations

- Sobel test allows to test whether there is any mediation effect in linear models, i.e.,

$$H_0 : NIE = 0 \iff M \text{ do not mediates the exposure effect}$$

- The test statistic is based on the asymptotic variance of the NIE OLS-based estimator, and is given by

$$Z_{stat} = \frac{\widehat{NIE}}{SE(\widehat{NIE})} = \frac{\hat{\alpha}_1 \hat{\beta}_2}{\widehat{var}(\hat{\beta}_2) \hat{\alpha}_1^2 + \hat{\beta}_2^2 \widehat{var}(\hat{\alpha}_1)}.$$

- That is distributed approximately $N(0, 1)$.
- Most statistical software has standard procedures to perform the Sobel test. However, it is straightforward to perform this test manually.
- Note: estimating the NIE by $\alpha_1 \hat{\beta}_2$ is also known as the product method (since we multiple the two coefficients/causal effects).

The models are

$$\mathbb{E}[M|X, L] = \alpha_0 + \alpha_1 X + \alpha_2 L, \quad \mathbb{E}[Y|X, M, L] = \beta_0 + \beta_1 X + \beta_2 M + \beta_3 L.$$

TE minus the direct effect

- Notably, by estimating

$$\mathbb{E}[Y|X, L] = \gamma_0 + \gamma_1 X + \gamma_2 L,$$

we obtain the total effect (TE) that is the estimator of γ_1 .

- Then, we can use the conditional outcome model to compute the NIE by

$$NIE = TE - NDE = \gamma_1 - \beta_1.$$

- This is known as the difference method.
- Numerically, we will obtain the exact same number as in the previous (product) method.
- By using the difference method, we don't need the mediator-exposure model, however, we still need to estimate two different models.

The models are

$$\mathbb{E}[M|X, L] = \alpha_0 + \alpha_1 X + \alpha_2 L, \quad \mathbb{E}[Y|X, M, L] = \beta_0 + \beta_1 X + \beta_2 M + \beta_3 L.$$

The total effect model

- The total effect model is just marginalized conditional outcome model over the mediator M

$$\mathbb{E}[\mathbb{E}[Y|X, L]|M] = \mathbb{E}[Y|X, L] = \gamma_0 + \gamma_1 X + \gamma_2 L.$$

- In linear models, the linearity of the total effect model is guaranteed once we assumed linear conditional outcome model.
- In non-linear models is the structure of the total effect model is not easily derived from the conditional outcome model, and needs to be assumed.
- Neither in linear nor non-linear models is the difference method generally more efficient than the multiplicative method. Furthermore, it may even add some more complexity to the computations.
- However, when the mediator-exposure model is severely misspecified, the difference method is preferable, as the total effect model and the conditional outcome model are generally more interpretable and more closely related.

- 1 Mediation analysis - DAG
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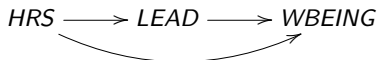
The variables and the causal structure

Bliese & Halverson (1996)

- Study by Bliese & Halverson (1996) examines the mediation effect of Leadership Climate (LEAD) on the the impact of work hours (HRS) on well-being (WBEING).
- WE will uses Sobel test to evaluate mediation.

Models' variables

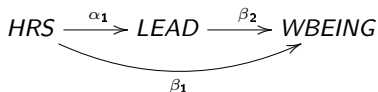
- Exposure: HRS (Work Hours) – Number of hours worked (numeric).
- Mediator: LEAD (Leadership Climate Perception) – Employees' collective perception of leadership quality (numeric).
- Outcome: WBEING (Well-Being) – Psychological well-being of employees (numeric).



Perception of leadership quality (LEAD) mediates the effect of work hours (HRS) on employees well being (WBEING)

Structural linear models

- Outcome (Direct-effect) model: $WBEING = \beta_0 + \beta_1 HRS + \beta_2 LEAD + \epsilon_Y$
- Exposure model: $LEAD = \alpha_0 + \alpha_1 HRS + \epsilon_M$
- Outcome (Total effect) model: $WBEING = \gamma_0 + \gamma_1 HRS + \tilde{\epsilon}_Y$



Perception of leadership quality (LEAD) mediates the effect of work hours (HRS) on employees well being (WBEING)

The natural direct and indirect effects

$$NDE = \beta_1, \quad NIE = \alpha_1 \beta_2, \quad TE = \gamma_1 = \beta_1 + \alpha_1 \beta_2.$$

Mediation Analysis results

For the presentation, we use the `sobel` function from the `multilevel` R-package.

```
$'Mod1: Y~X'
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	3.51693620	0.052902697	66.47934	0.000000e+00
pred	-0.06523285	0.004590274	-14.21110	3.078129e-45

```
$'Mod2: Y~X+M'
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.86832973	0.06413083	29.13310	1.024201e-176
pred	-0.04311316	0.00421918	-10.21837	2.382257e-24
med	0.48386196	0.01242129	38.95426	4.967825e-302

```
$'Mod3: M~X'
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	3.40718349	0.045154735	75.45573	0.000000e+00
pred	-0.04571488	0.003917997	-11.66792	3.488366e-31

```
$z.value
```

```
[1] -11.17729
```

Testing the null hypothesis of no mediation effect

- The estimate natural indirect effect (NIE) is:

$$\hat{\alpha}_1\hat{\beta}_2 = 0.48386196 \times -0.04571488 = -0.0221$$

- The computed standard error of the NIE is 0.00198, thus the Z-statistic of the Sobel test is

$$Z_{stat} = -11.17729$$

with a $p.value < 0.001$.

- Namely, we reject the null hypothesis $H_0 : NIE = 0$, and conclude that the perception of leadership qualities indeed mediate the causal effect of work hours on the employees well being.

Direct and total effects

- The estimated natural direct effect (NDE) of the work hours on employees well being is

$$\hat{\beta}_1 = -0.04311316$$

- The estimated total effect (TE) of the work hours on employees well being is

$$\hat{\gamma}_1 = -0.06523285 = \hat{\beta}_1 + \hat{\alpha}_1\hat{\beta}_2 = -0.04311316 - 0.0221$$

- Increase in work hours (HRS) negatively impact employees' perception of leadership quality (LEAD).
- Poor perceived leadership quality (LEAD) further reduces well-being (WBEING).
- The effect of work hours on well-being is partially (about 30%) mediated by leadership climate.

Results

- Longer work hours cause worse attitude towards the leadership quality.
- Poor leadership quality perception lowers employees' well-being.
- Fixing work hours alone may not improve well-being; improving leadership quality can buffer stress and mitigate the total negative effect of workload on well-being.

Practical implications

- Leadership climate (perceived leadership quality) plays a critical role in employee well-being.
- Organizational interventions should focus not just on reducing workload but also improving leadership quality (or at least the perception of the leadership quality).

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References

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